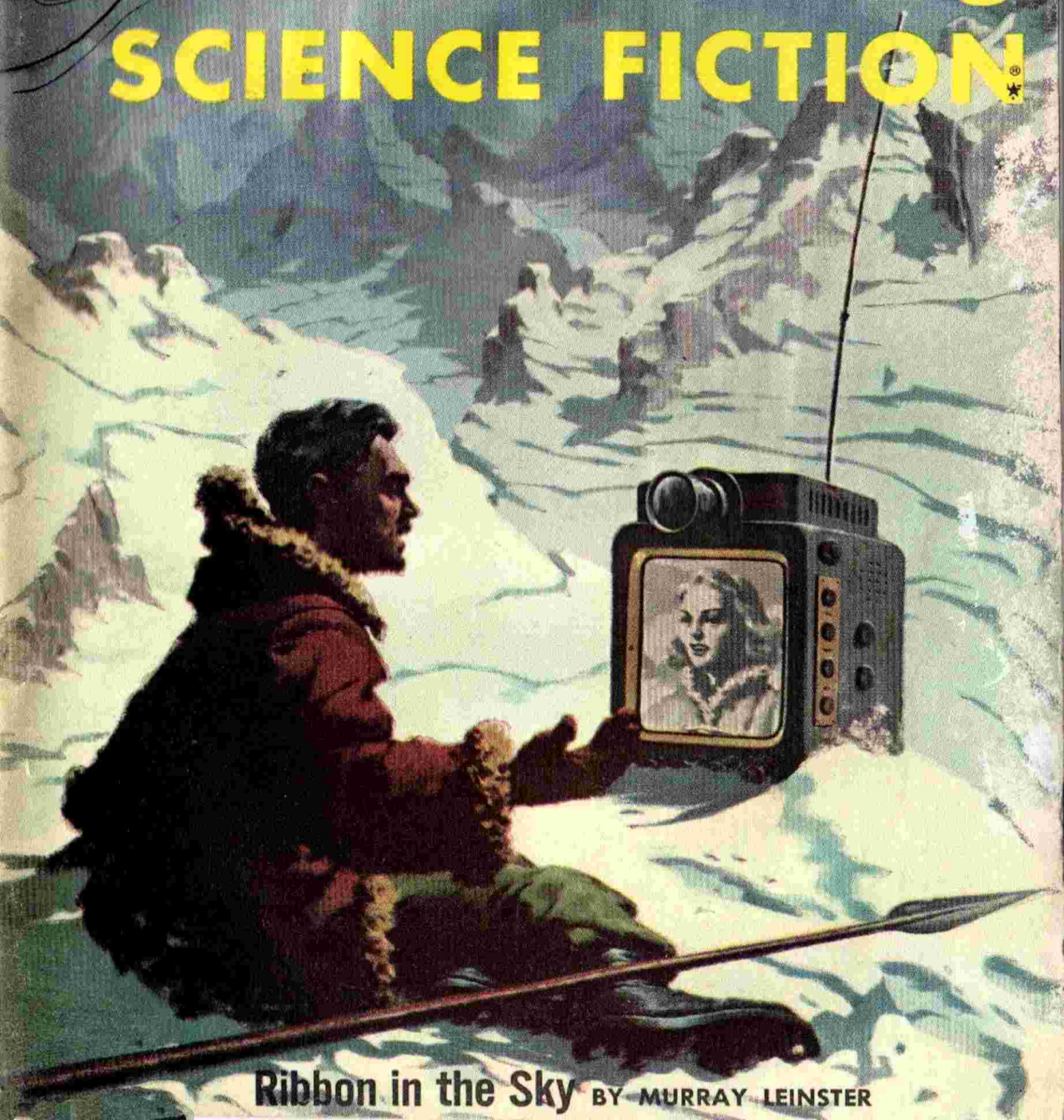


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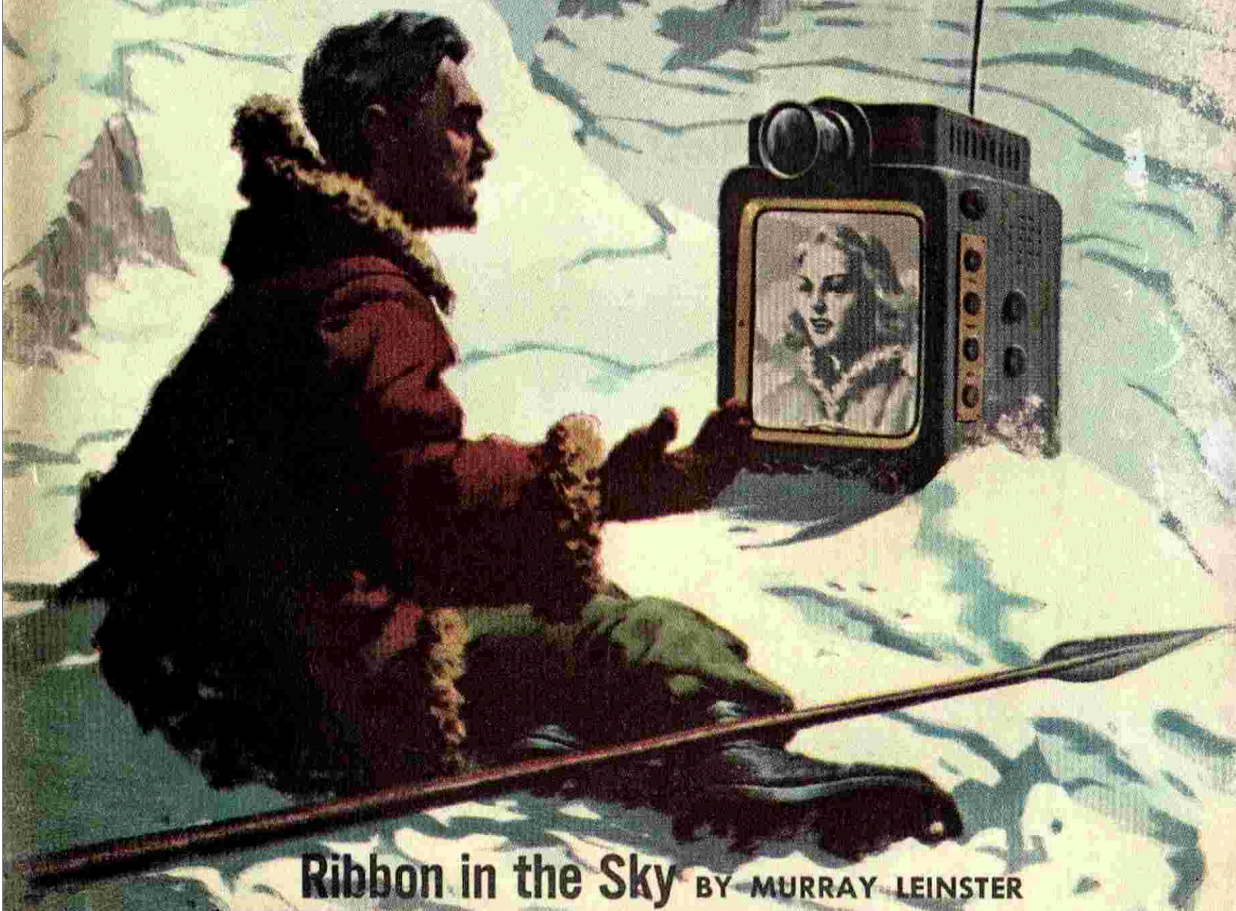
Astounding SCIENCE FICTION®



Ribbon in the Sky BY MURRAY LEINSTER

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SKY ***

RIBBON IN THE SKY

BY MURRAY LEINSTER

Illustrated by van Dongen

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I

"An error is a denial of reality, but mistakes are mere mental malfunctionings. In an emergency, a mistake may be made because of the need for precipitate action. There is no time to choose the best course: something must be done at once. Most mistakes, however, are made without any such exterior pressure. One accepts the first-imagined solution to a problem without examining it, either out of an urgent desire to avoid the labor of thinking, or out of impassioned reluctance to think about the matter at hand when prettier and more pleasurable other things can be contemplated...."

The Practice of Thinking
Fitzgerald

It turned out afterward that somebody had punched the wrong button in a computer. It was in a matter in which mistakes are not permissible, but just as nothing can be manufactured without an ordinary hammer figuring somewhere in the making or the making-ready-to-make, so nothing can be done without a fallible human operating at some stage of the proceedings. And humans make mistakes casually, off-handedly, with impartial lack of malice, and unpredictability. So....

Calhoun heard the tape-speaker say, "*When the gong sounds, breakout will follow in five seconds.*" Then it made solemn ticking noises while Calhoun yawned and put aside the book, "The Practice of Thinking," that he'd been studying. Study was a necessity in his profession. Besides, it helped to pass the time in overdrive. He went to the control-desk chair and strapped in. Murgatroyd the *tormal* uncoiled his tail from about his nose and stood up from where he was catching twenty winks. He padded to the place under Calhoun's chair where there were things to grab hold of, if necessary, with four black paws and a prehensile tail.

"*Chee*," said Murgatroyd conversationally in his shrill treble.

"I agree," Calhoun told him gravely. "Stone walls do not a prison make, nor Med Ship hulls a cage. But it will be good to get outside for a change."

The tape-speaker ticked and tocked and ticked and tocked. There was the sound of a gong. A voice said measuredly, "*Five—four—three—two—one* —"

The ship came out of overdrive. Calhoun winced and swallowed. Nobody ever gets used to going into overdrive or coming out of it. One is hideously dizzy for an instant, and his stomach has a brief but violent urge to upchuck, and no matter how often one has experienced it, it is necessary to fight a flash of irrational panic caused by the two sensations together.

But after an instant Calhoun stared about him as the vision-screens came to life. They showed the cosmos outside the Med Ship. It was a perfectly normal cosmos—not at all the cosmos of overdrive—but it looked extremely wrong to Calhoun. He and Murgatroyd and the Med Ship were in emptiness. There were stars on every hand, and they were of every conceivable color and degree of brightness. But every one of them was a point of light, and a point only.

This, obviously, was not what he'd expected. These days ships do not stop to view the universe from the monstrous loneliness which is Between-the-Stars. All ships go into overdrive as near their port of departure as they can. Usually it is something like five or six planetary diameters out from the local spaceport. All ships come out of overdrive as near their destinations as computation makes possible. They do not stop to look at scenery on the way. It isn't good for humans to look at stars when there are only stars to see. The sight has a tendency to make them feel small. Too small. Men have been known to come out of such an experience gibbering.

Calhoun scowled at the sight of Between-the-Stars. This was not good. But he wasn't frightened—not yet. There should have been a flaming sun somewhere nearby, and there should have been bright crescents or half-disks or mottled cloudy planets swimming within view. The sun should have been the star Merida, and Calhoun should land in commonplace fashion on Merida II and make a routine planetary health check on a settled, complacent population, and presently he should head back to Med Headquarters with a report containing absolutely nothing of importance.

But he couldn't do any of these things. He was in purely empty space. It was appalling.

Murgatroyd jumped up to the arm of the control-chair, to gaze wisely at the screens. Calhoun continued to scowl. Murgatroyd imitated him with a *tormal's* fine complacency in duplicating a man's actions. What he saw meant nothing to him, of course. But he was moved to comment.

"*Chee*," he said shrilly.

"To be sure," agreed Calhoun distastefully. "That is a very sage observation, Murgatroyd. But I deplore the situation that calls for it. Somebody's bilged on us."

Murgatroyd liked to think that he was carrying on a conversation. He said zestfully:

"*Chee-chee! Chee-chee-chee!*"

"No doubt," conceded Calhoun. "But this is a mess! Hop down and let me try to get out of it."

Murgatroyd disappointedly hopped to the floor. He watched with bright eyes as Calhoun annoyedly went to the emergency-equipment locker and brought out the apparatus designed to take care of a state of things like this. If the situation wasn't too bad, correcting it should be simple enough. If it was too bad, it could be fatal.

The average separation of stars throughout the galaxy, of course, is something like four or five light-years. The distance between sol-type stars is on an average very much higher, and with certain specific exceptions habitable planets are satellites of sol-type suns. But only a fraction of the habitable planets are colonized, and when a ship has traveled blind, in overdrive, for two months or more its pilot cannot simply look astern and recognize his point of departure. There's too much scenery in between. Further, nobody can locate himself by the use of star-maps unless he knows where something on the star-map is, with reference to himself. Which makes a star-map not always useful.

But the present blunder might not be serious. If the Med Ship had come out into normal space no more than eight to ten light-years from Merida, Calhoun might identify that sun by producing parallax. He could detect relative distances for a much greater range. But it was to be hoped that his present blunder was small.

He got out the camera with its six lenses for the six vision-screens which showed space in all directions. He clamped it in place and painstakingly snapped a plate. In seconds he had everything above third magnitude faithfully recorded in its own color, and with relative brightnesses expressed in the size of the dots of tint. He put the plate aside and said:

"Overdrive coming, Murgatroyd."

He pressed the short-hop button and there was dizziness and nausea and a flash of fear—all three sensations momentary. Murgatroyd said, "*Chee*" in a protesting tone, but Calhoun held down the button for an accurate five minutes. He and Murgatroyd gulped together when he let up the button again and all space whirled and nausea hit as before. He took another plate of all the heavens, made into one by the six-lensed camera. He swung the ship by ninety degrees and pressed the short-hop button a second time. More dizziness and panic and digestive revolt. In five minutes it was repeated as the ship came out to normal space yet again.

"*Chee-chee!*" protested Murgatroyd. His furry paws held his round little belly against further insult.

"I agree," said Calhoun. "I don't like it either. But I want to know where we are—if anywhere."

He set up the comparator and inserted the three plates. Each had images of each of the six vision-screens. When the instrument whirled, each of the plates in turn was visible for part of a second. Extremely remote stars would not jiggle perceptibly—would not show parallax—but anything within twenty light-years should. The jiggling distance could be increased by taking the plates still farther apart. This time, though, there was one star which visibly wavered in the comparator. Calhoun regarded it suspiciously.

"We're Heaven knows where," he said dourly. "Somebody really messed us up! The only star that shows parallax isn't Merida. In fact, I don't believe in

it at all! Two plates show it as a sol-class sun and the third says it's a red dwarf!"

On the face of it, such a thing was impossible. A sun cannot be one color as seen from one spot, and another color seen from another. Especially when the shift of angle is small.

Calhoun made rough computations. He hand-set the overdrive for something over an hour's run in the direction of the one star-image which wobbled and thereby beckoned. He threw the switch. He gulped, and Murgatroyd acted for a moment as if he intended to yield unreservedly to the nausea of entering overdrive. But he refrained.

There was nothing to do but kill time for an hour. There was a micro-reel of starplates, showing the heavens as photographed with the same galactic coordinates from every visited sol-class star in this sector of the galaxy. Fewer than one in forty had a colonized planet, but if the nearest had been visited before, and if the heavens had been photographed there, by matching the stars to the appropriate plate he could find out where he was. Then a star-map might begin to be of some use to him. But he had still to determine whether the error was in his astrogation unit, or in the data fed to it. If the first, he'd be very bad off indeed. If the second, he could still be in a fix. But there was no point in worrying while in overdrive. He lay down on his bunk and tried to concentrate again on the book he'd laid aside.

"Human error, moreover," he read, "is never purely random. The mind tends to regard stored data as infallible and to disregard new data which contradicts it..." He yawned, and skipped. *"... So each person has a personal factor of error which is not only quantitative but qualitative...."*

He read on and on, only half absorbing what he read. But a man who has reached the status of a Med Ship man in the Interstellar Medical Service hasn't finished learning. He's still away down the ladder of rank. He has plenty of studying ahead of him before he gets very far.

The tape-speaker said, *"When the gong sounds, breakout will be five seconds off."* It began to *ticktock*, slowly and deliberately. Calhoun got into

the control seat and strapped in. Murgatroyd said peevishly, "*Chee!*" and went to position underneath the chair. The voice said, "*Five—four—three—two—one—*"

The little Med Ship came out of overdrive, and instantly its emergency rockets kicked violently and Murgatroyd held desperately fast. Then the rockets went off. There'd been something unguessable nearby—perhaps cometary debris at the extremest outer limit of a highly eccentric orbit. Now there was a starfield and a sun within two light-hours. But if Calhoun had stared, earlier, when there was no sun in sight at all, now he gazed blankly at the spectacle before him.

There was a sun off to starboard. It was a yellow sun—a sol-type star with a barely perceptible disk. There were planets. Calhoun saw immediately one gas-giant near enough to be more than a point, and a sliver of light which was the crescent of another more nearly in line toward the sun. But he gazed at a belt, a band, a ribbon of shining stuff which was starkly out of all reason.

It was a thin curtain of luminosity circling this yellow star. It was not a ring from the breakup of a satellite within Roche's Limit. There were two quite solid planets inside it and nearer to the star. It was a thin, wide, luminous golden ribbon which looked like something that needed a flatiron to smooth it out. It looked something like an incandescent smoke ring. It was not smooth. It had lumps in it. There were corrugations in it. An unimaginable rocket with a flat exhaust could have made it while chasing its tail around the sun. But that couldn't have happened, either.

Calhoun stared for seconds.

"Now," he said, "I've seen everything!" Then he grunted as realization came. "And we're all right, Murgatroyd! It's not our computers that went wrong. Somebody fed them wrong data. We arrived where we aimed for, and there'll be a colonized planet somewhere around."

He unlimbered the electron telescope and began a search. But he couldn't resist a closer look at the ribbon in space. It had exactly the structure of a slightly wobbly wrinkled ribbon without beginning or end. It had to be a complex of solid particles, of course, and an organization of solid particles cannot exist in space without orbital motion. But orbits would smooth out in

the course of thousands of revolutions around a primary. This was not smoothed out. It was relatively new.

"It's sodium dust," said Calhoun appreciatively. "Or maybe potassium. Hung out there on purpose. Particles small enough to have terrific surface and reflective power, and big enough not to be pushed out of orbit by light-pressure. Clever, Murgatroyd! At a guess it'll have been put out to take care of the climate on a planet just inside it. Which would be—there! Let's go look!"

He was so absorbed in his admiration that the almost momentary overdrive-hop needed for approach went nearly unnoticed. He even realized—his appreciation increasing—that this cloud of tiny particles accounted for the red-dwarf appearance on one of the plates he'd taken. Light passing through widely dispersed and very small particles turns red. From one position, he'd photographed through this dust cloud.

The ribbon was a magnificent idea—the more magnificent because of its simplicity. It would reflect back otherwise wasted sun-heat to a too-cold planet and make it warmer. There was probably only an infinitesimal actual mass of powder in the ring, at that. Tens or scores of tons in all. Hardly more.

The planet for which it had been established was the third world out. As is usual with sol-class systems, the third planet's distance from the sun was about a hundred twenty million miles. It had icecaps covering more than two-thirds of its surface. The sprawling white fingers of glaciation marked mountain chains and highlands nearly to the equator. But there was some blue sea, and there was green vegetation in a narrow belt of tropicity.

Calhoun jockeyed the Med Ship to position for a landing call. This was not Merida II, but there should be a colony here. That glowing ribbon had not been hung out for nothing.

"Med Ship Esclipus Twenty," he said confidently into the spacephone mike. "Calling ground. Requesting co-ordinates for landing. My mass is fifty tons. Repeat, five-oh tons. Purpose of landing, to find out where I am and how to get where I belong."

There was a clicking. Calhoun repeated the call. He heard murmurings which were not directed into the transmitter on the planet. He heard an

agitated, "*How long since a ship landed?*" Another voice was saying fiercely, "*Even if he doesn't come from Two City or Three City, who knows what sickness—*" There was sudden silence, as if a hand had been clapped over the microphone below. Then a long pause. Calhoun made the standard call for the third time.

"*Med Ship Esclipus Twenty,*" said the spacephone speaker grudgingly, "*You will be allowed to land. Take position—*" Calhoun blinked at the instructions he received. The co-ordinates were not the normal galactic ones. They gave the local time at the spaceport, and the planetary latitude. He was to place himself overhead. He could do it, of course, but the instructions were unthinkable. Galactic co-ordinates had been used ever since Calhoun knew anything about such matters. But he acknowledged the instructions. Then the voice from the speaker said truculently: "*Don't hurry! We might change our minds! And we have to figure settings for an only fifty-ton ship, anyhow.*"

Calhoun's mouth dropped open. A Med Ship was welcome everywhere, these days. The Interstellar Medical Service was one of those overworked, understaffed, kicked-around organizations which is everywhere taken for granted. Like breathable air, nobody thought to be grateful for it—but nobody was suspicious of it, either.

The suspicion and the weird co-ordinates and the ribbon in space combined to give Calhoun a highly improbable suspicion. He looked forward with great interest to this landing. He had not been ordered to land here, but he suspected that a Med Ship landing was a long, long time overdue.

"I forgot to take star-pictures," he told Murgatroyd, "but a ribbon like this would have been talked about if it had been reported before. I doubt star-pictures would do us any good. The odds are our only chance to find out where we are is to ask." Then he shrugged his shoulders. "Anyhow this won't be routine!"

"*Chee!*" agreed Murgatroyd, profoundly.

II

"An unsolvable but urgent problem may produce in a society, as in an individual, an uncontrollable emotional tantrum, an emotional denial of the problem's existence, or purposive research for a solution. In olden days, the first reaction produced mass-tantrums then called 'wars.' The second produced frenziedly dogmatic ideologies. The third produced modern civilization. All three reactions still appear in individuals. If the first two should return to societies, as such...."

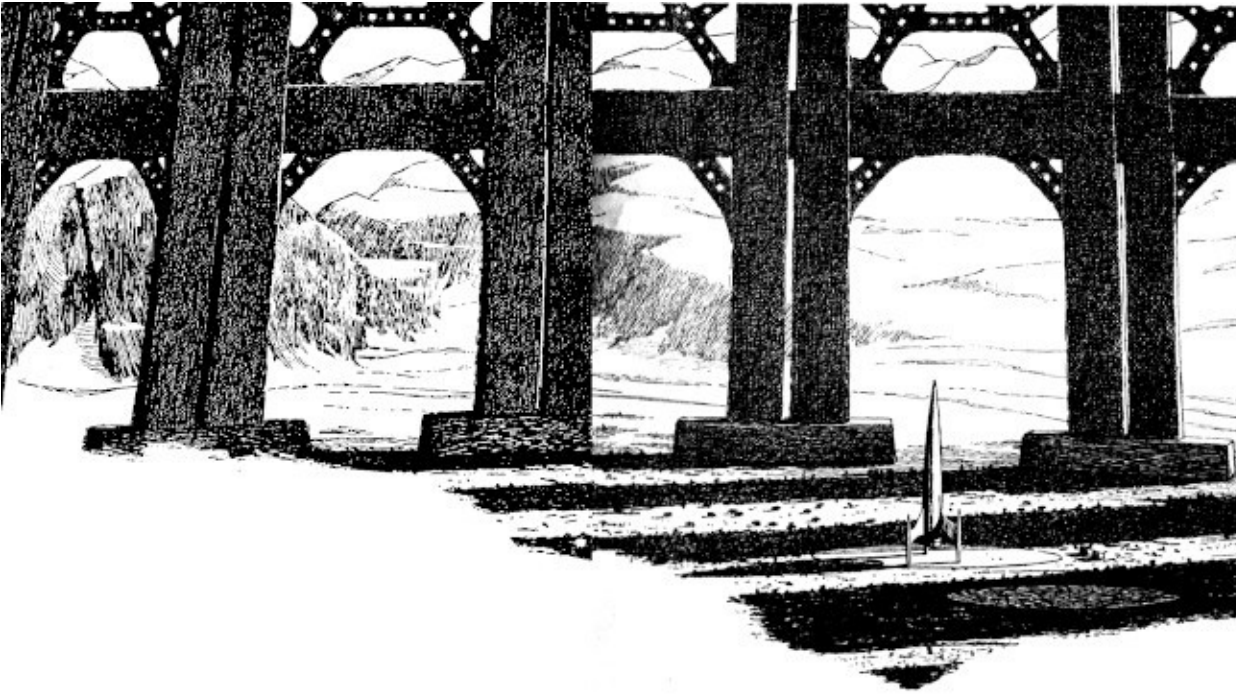
The Practice of Thinking
Fitzgerald

The descent, at least, was not routine. It was nerve-racking. The force-field from the planet's giant steel landing grid reached out into space and fumbled for the Med Ship. That was clumsily done. When it found the ship, it locked on. And that was awkwardly handled. The rest was worse. Whoever handled the controls, aground, was hopelessly inept. Once the Med Ship's hull-temperature began to climb, and Calhoun had to throw on the spacephone and yelp for caution. He did not see as much of the nearing planet as he'd have liked.

At fifty miles of height, the last trace of blue sea vanished around the bulge of the world. At twenty miles, the mountain chains were clearly visible, with their tortured, winding ice rivers which were glaciers. At this height three patches of green were visible from aloft. One, directly below, was little more than a mile in diameter and the landing grid was its center and almost its circumference. Another was streaky and long, and there seemed to be heavy mist boiling about it and above it. The third was roughly triangular. They were many miles apart. Two of them vanished behind mountains as the ship descended.

There were no cities in view. There were no highways. This was an ice world with bare ground and open water at its equator only. The spaceport was placed in a snow-ringed polar valley.

Near landing, Calhoun strapped in because of the awkwardness with which the ship was lowered. He took Murgatroyd on his lap. The small craft bounced and wobbled as unskilled hands let it down. Presently, Calhoun saw the angular girders of the landing grid's latticed top rise past the opened ports. Seconds later, the Med Ship bumped and slid and bounced heart-stoppingly. Then it struck ground with a violent jolt.



Calhoun got his breath back as the little ship creaked and adjusted itself to rest on its landing fins after some months in space.

"Now," said the voice in the spacephone speaker—but it sounded as if it were trying to conceal relief—"now stay in your ship. Our weapons are bearing on you. You may not come out until we've decided what to do about you."

Calhoun raised his eyebrows. This was very unusual indeed. He glanced at the external field indicator. The landing grid field was off. So the operator bluffed. In case of need Calhoun could blast off on emergency rockets and probably escape close-range weapons anyhow—if there were any—and he could certainly get around the bulge of the world before the amateur at the grid's controls could hook on to him again.

"Take your time," he said with irony. "I'll twiddle my fingers. I've nothing better to do!"

He freed himself from his chair and went to a port to see. He regarded the landscape about him with something like unbelief.

The landing grid itself was a full mile across and half as high. It was a vast, circular frame of steel beams reaching heavenward, with the curiously curving copper cables strung as they had to be to create the highly special force-field which made space transportation practical. Normally such gigantic structures rose in the centers of spaceport cities. They drew upon the planet's ionosphere for power to lift and land cargo ships from the stars, and between-times they supplied energy for manufactures and the operation of cities. They were built, necessarily, upon stable bedrock formations, and for convenience were usually located where the cargoes to be shipped would require least surface transportation.

But here there was no city. There was perhaps a thousand acres of greenness—a mere vague rim around the outside of the grid. There was a control-room building to one side, of course. It was solidly built of stone, but there had been an agglomeration of lean-tos added to it with slanting walls and roofs of thin stratified rock. And there were cattle grazing on the green grass. The center of the grid was a pasture!

Save for the clutter about the grid-control building there were no structures, no dwellings, no houses or homes anywhere in view. There was no longer even a highway leading to the grid. Calhoun threw on the outside microphones and there was no sound except a thin keening of wind in the steelwork overhead. But presently one of the cattle made a mournful bellowing sound.

Calhoun whistled as he went from one port to another.

"Murgatroyd," he said meditatively on his second round, "you observe—if you observe—one of the consequences of human error. I still don't know where I am, because I doubt that starplates have ever been made from this solar system, and I didn't take one for comparison anyhow. But I can tell you that this planet formerly had a habitability rating of something like oh point oh, meaning that if somebody wanted to live here it would be possible but it wouldn't be sensible. But people did come here, and it was a mistake."

He stared at a human figure, far away. It was a woman, dressed in shapeless, badly draping garments. She moved toward a clump of dark-coated cattle and did something in their midst.

"The mistake looks pretty evident to me," added Calhoun, "and I see some possibilities I don't like at all. There is such a thing as an isolation syndrome, Murgatroyd. A syndrome is a complex of pathological symptoms which occur together as a result of some morbid condition. To us humans, isolation is morbid. You help me to endure it, Murgatroyd, but I couldn't get along with only your society—charming as it is—for but so long. A group of people can get along longer than a single man, but there is a limit for any small-sized group."

"*Chee*," said Murgatroyd.

"In fact," said Calhoun, frowning, "there's a specific health problem involved, which the Med Service recognizes. There can be partial immunity, but there can be some tricky variations. If we're up against a really typical case we have a job on hand. And how did these people get that dust-ring out in space? They surely didn't hang it out themselves!"

He sat down and scowled at his thoughts. Presently he rose again and once more surveyed the icy landscape. The curious green pasture about the landing grid was highly improbable. He saw glaciers over-hanging this valley. They were giant ice rivers which should continue to flow and overwhelm this relatively sheltered spot. They didn't. Why not?

It was more than an hour before the spacephone clattered. When Calhoun threw the switch again a new voice came out of it. This was also a male voice, but it was high-pitched as if from tension.

"*We've been talking about you*," said the voice. It quivered with agitation which was quite out of reason. "*You say you're Med Service. All right. Suppose you prove it!*"

The landed Med Ship should be proof enough for anybody. But Calhoun said politely:

"I have the regular identifications. If you'll go on vision, I'll show you my credentials."

"Our screen's broken," said the voice, suspiciously, "but we have a sick cow. It was dumped on us night before last. Cure her and we'll accept it as identification."

Calhoun could hardly believe his ears. This was an emergency situation! The curing of a sick cow was considered more convincing than a Med Ship man's regular credentials! Such a scale of values hinted at more than a mere isolation syndrome. There were thousands of inhabited worlds, now, with splendid cities and technologies which most men accepted with the same bland confidence with which they looked for sunrise. The human race was civilized. Suspicion of a Med Ship was unheard of. But here was a world—

"Why ... certainly," said Calhoun blankly. "I suppose I may go outside to ... ah ... visit the patient?"

"We'll drive her up to your ship," said the high, tense voice. "And you stay close to it!" Then it said darkly. "Men from Two City sneaked past our sentries to dump it on us. They want to wipe out our herd! What kind of weapons have you got?"

"This is a Med Ship!" protested Calhoun. "I've nothing more than I might need in an emergency!"

"We'll want them anyhow," said the voice. "You said you need to find out where you are. We'll tell you, if you've got enough weapons to make it worth while."

Calhoun drew a deep breath.

"We can argue that later," he said. "I'm just a trifle puzzled. But first things first. Drive your cow."

He held his head in his hands. He remembered to throw off the spacephone and said:

"Murgatroyd, say something sensible! I never ran into anybody quite as close to coming apart at the seams as that! Not lately! Say something rational!"

Murgatroyd said, *"Chee?"* in an inquiring tone.

"Thanks," said Calhoun. "Thanks a lot."

He went back to the ports to watch. He saw men come out of the peculiar agglomeration of buildings that had been piled around the grid's sturdy control building. They were clothed in cloth that was heavy and very stiff, to judge by the way it shifted with its wearers' movements. Calhoun wasn't familiar with it. The men moved stolidly, on foot, across the incredible pasture which had been a landing space for ships of space at some time or other.

They reached a spot where a dark animal form rested on the ground. Calhoun hadn't noticed it particularly. Cattle, he knew, folded their legs and lay down and chewed cuds. They existed nearly everywhere that human colonies had been built. On some worlds there were other domestic animals descended from those of Earth. Of course there were edible plants and some wholesome animals which had no connection at all with humanity's remote ancestral home, but from the beginning human beings had been adjusted to symbiosis with the organic life of Earth. Foodstuffs of non-terrestrial origin could supplement Earth-food, of course. In some cases Earth-foods were the supplements and local, non-terrestrial foodstuffs the staples. But human beings did not thrive on a wholly un-Earthly diet.

The clump of slowly moving men reached the reclining cow. They pulled up stakes which surrounded her, and coiled up wire or cordage which had made the stakes into a fence. They prodded the animal. Presently it lurched to its feet and swung its head about foolishly. They drove it toward the Med Ship.

Fifty yards away they stopped, and the outside microphones brought the sound of their voices muttering. By then Calhoun had seen their faces. Four of the six were bearded. The other two were young men. On most worlds men prided themselves that they needed to shave, but few of them omitted the practice.

These six moved hastily away, though the two younger ones turned often to look back. The cow, deserted, stumbled to a reclining position. It lay down, staring stupidly about. It rested its head on the ground.

"I go out now, eh?" asked Calhoun mildly.

"We're watching you!" grated the spacephone speaker.

Calhoun glanced at the outside temperature indicator and added a garment. He put a blaster in his pocket. He went out the exit port.

The air was bitter cold, after two months in a heat-metered ship, but Calhoun did not feel cold. It took him seconds to understand why. It was that the ground was warm. Radiant heat kept him comfortable, though the air was icy. Heat elements underground must draw power from somewhere—the grid's tapping of the ionosphere—and heated this pasture from underneath so forage plants could grow here. They did. The cattle fed on them. There would be hydroponic gardens somewhere else, probably underground. They would supply vegetable food in greater quantity. But in the nature of things human beings had to have animal food in a cold climate.

Calhoun went across the pasture with the frowning snowy mountains all about. He regarded the reclining beast with an almost humorous attention. He did not know anything about the special diseases of domestic animals. He had only the knowledge required of a Med Ship man. But that should be adequate. The tense voice had said that this beast had been "dumped," to "wipe out" the local herd. So there would be infection and there would be some infective agent.

He painstakingly took samples of blood and saliva. In a ruminant, certainly, any digestive-tract infection should show up in the saliva. He reflected that he did not know the normal bovine temperature, so he couldn't check it. Nor the respiration. But the Interstellar Medical Service was not often called on to treat ailing cows.

Back in the ship he diluted his samples and put droplets in the usual nutrient solutions. He sealed up droplets in those tiny slides which allow a culture to be examined as it grows. His microscope, of course, allowed of inspection under light of any wave length desired, and so yielded information by the frequency of the light which gave clearest images of different features of microorganisms.

After five minutes of inspection he grunted and hauled out his antibiotic stores. He added infinitesimal traces of cillin to the culture-media. In the

microscope, he watched the active microscopic creatures die. He checked with the other samples.

He went out to the listless, enfeebled animal. He made a wry guess at its body-weight. He used the injector. He went back to the Med Ship. He called on the spacephone.

"I think," he said politely, "that your beast will be all right in thirty hours or so. Now, how about telling me the name of this sun?"

The voice said sharply:

"There's a matter of weapons, too! Wait till we see how the cow does! Sunset will come in an hour. When day comes again, if the cow is better—we'll see!"

There was a click. The spacephone cut off.

Calhoun pulled out the log-mike. There was already an audio record of all ship-operations and communications. Now he added comments—a description of the ribbon in the sky, the appearance of the planet, and such conclusions as he'd come to. He ended:

"... The samples from the cow were full of a single coccus, which seemed to have no resistance to standard antibiotics. I pumped the beast full of cillin and called it a day. I'm concerned, though, because of the clear signs of an isolation syndrome here. They're idiotically suspicious of me and won't even promise a bargain, as if I could somehow overreach them because I'm a stranger. They've sentries out—they said somebody sneaked past them—against what I imagine must be Two City and Three City. I've an impression that the sentries are to enforce a quarantine rather than to put up a fight. It is probable that the other communities practice the same tactics—plus biological cold war if somebody did bring a sick cow here to infect and destroy the local herd. These people may have a landing grid, but they've an isolation syndrome and I'm afraid there's a classic Crusoe health problem in being. If that's so, it's going to be nasty!"

He cut off the log. The classic Crusoe problem would be extremely awkward if he'd run into it. There was a legend about an individual back on old Earth who'd been left isolated on an island by shipwreck for half a lifetime. His

name was given to the public-health difficulties which occurred when accidental isolations occurred during the chaotic first centuries of galactic migration. There was one shipwreck to which the name was first applied. The ship was missing, and the descendants of the crew and passengers were not contacted until three generations had passed. Larger-scale and worse cases occurred later, when colonies were established by entrepreneurs who grew rich in the establishment of the new settlements, and had no interest in maintaining them. Such events could hardly happen now, of course, but even a Crusoe condition was still possible in theory. It might exist here. Calhoun hoped not.

It did not occur to him that the affair was not his business because he hadn't been assigned to it. He belonged to the Med Service, and the physical well-being of humans everywhere was the concern of that service. If people lived by choice in an inhospitable environment, that was not a Med man's problem, but anything which led to preventable deaths was. And in a Crusoe colony there were plenty of preventable deaths!

He cooked a meal to have something to occupy his mind. Murgatroyd sat on his haunches and sniffed blissfully. Presently Calhoun ate, and again presently darkness fell on this part of the world. There were new noises—small ones. He went to look. The pasture inside the landing grid was faintly lighted by the glowing ribbon in the sky. It looked like a many-times-brighter Milky Way. The girders of the landing grid looked very black against it.

He saw a dark figure plodding away until he vanished. Then he reappeared as a deeper black against the snow beyond the pasture. He went on and on until he disappeared again. A long time later another figure appeared where he'd gone out of sight. It plodded back toward the grid. It was a different individual. Calhoun had watched a changing of sentries. Suspicion. Hostility. The least attractive qualities of the human race, brought out by isolation.

There could not be a large population here, since such suspicions existed. And it was divided into—most likely—three again-isolated communities. This one had the landing grid, which meant power, and a spacephone but no vision screen attached to it. The fact that there were hostile separate communities made the situation much more difficult, from a medical point of view. It multiplied the possible ghastly features which could exist.

Murgatroyd ate until his furry belly was round as a ball, and settled to stuffed slumber with his tail curled around his nose. Calhoun tried to read. But he was restless. His own time-cycle on the ship did not in the least agree with the time of daylight on this planet. He was wakeful when there was utter quiet outside. Once one of the cattle made a dismal lowing noise. Twice or three times he heard cracking sounds, like sharp detonations, from the mountains. They would be stirrings in the glaciers.

He tried to study, but painstaking analysis of the methods by which human brains defeated their own ends and came up with wrong answers was not appealing. He grew horribly restless.

It had been dark for hours when he heard rustling noises on the ground outside—through the microphones, of course. He turned up the amplification and made sure that a small party of men moved toward the Med Ship. From time to time they paused, as if in caution.

"Murgatroyd," he said dryly, "we're going to have visitors. They didn't give notice by spacephone, so they're unauthorized."

Murgatroyd blinked awake. He watched as Calhoun made sure of the blaster in his pocket and turned on the log-mike. He said:

"All set, Murgatroyd?"

Murgatroyd said "*Chee*" in his small shrill voice just as a soft and urgent knock sounded on the exit-lock door. It was made with bare knuckles. Calhoun grimaced and went into the lock. He undogged the door and began to open it, when it was whipped from his grasp and plunging figures pushed in. They swept him back into the Med Ship's cabin. He heard the lock-door close softly. Then he faced five roughly, heavily clothed men who wore cloaks and mittens and hoods, with cloth stretched tightly across their faces below the eyes. He saw knives, but no blasters.

A stocky figure with cold gray eyes appeared as spokesman.

"You're the man who got landed today," he said in a deep voice and with an effect of curtness. "My name's Hunt. Two City. You're a Med Ship man?"

"That's right," said Calhoun. The eyes upon him were more scared than threatening—all but the stocky man named Hunt. "I landed to find out where I was," he added. "The data-card for my astrogator had been punched wrong. What—"

"You know about sickness, eh?" demanded the stocky man evenly. "How to cure it and stop it?"

"I'm a Med Ship man," admitted Calhoun. "For whatever that may mean."

"You're needed in Two City," said the deep-voiced Hunt. His manner was purest resolution. "We came to get you. Get y'medicines. Dress warm. Load us down, if you like, with what you want to take. We got a sledge waiting."

Calhoun felt a momentary relief. This might make his job vastly easier. When isolation and fear brings a freezing of the mind against any novelty—even hope—a medical man has his troubles. But if one community welcomed him—

"*Chee!*" said Murgatroyd indignantly from overhead. Calhoun glanced up and Murgatroyd glared from a paw-hold near the ceiling. He was a peaceable animal. When there was scuffling he got out of the way. But now he chattered angrily. The masked men looked at him fearfully. But their deep-voiced leader growled at them.

"Just a animal." He swung back to Calhoun. "We got a need for you," he repeated. "We mean all right, and anything we got you can have if you want it. But you're coming with us!"

"Are your good intentions," asked Calhoun, "proved by your wearing masks?"

"They're to keep from catchin' your sickness," said the deep voice impatiently. "Point to what you want us to take!"

Calhoun's feeling of encouragement vanished. He winced a little. The isolation syndrome was fully developed. It was a matter of faith that strangers were dangerous. All men were assumed to carry contagion. Once, they'd have been believed to carry bad luck. But a regained primitiveness would still retain some trace of the culture from which it had fallen. If there were three settlements as the pastureslands seen from space suggested, they would not believe in magic, but they would believe in contagion. They might have, or once have had, good reason. Anyhow they would fanatically refrain

from contact with any but their own fellow-citizens. Yet there would always be troubles to excite their terrors. In groups of more than a very few there would always be an impulse against the isolation which seemed the only possible safety in a hostile world. The effectiveness of the counter-instinct would depend in part on communications, but the urge to exogamy can produce ghastly results in a small culture gone fanatic.

"I think," said Calhoun, "that I'd better come with you. But the people here have to know I've gone. I wouldn't like them to heave my ship out to space in pure panic because I didn't answer from inside it!"

"Leave a writing," said Hunt's deep voice, as impatiently as before. "I'll write it. Make them boil, but they don't dare follow us!"

"No?"

"Think One City men," asked the stocky man scornfully, "will risk us toppling avalanches on them?"

Calhoun saw. Amid mountain-country in a polar zone, travel would be difficult at best. These intruders had risked much to come here for him. But they were proud of their daring. They did not believe that the folk of lesser cities—tribes—groups than theirs had courage like theirs. Calhoun recognized it as a part of that complex of symptoms which can begin with an epidemic and end with group-madness.

"I'll want this—and this—and that," said Calhoun. He wouldn't risk his microscope. Antibiotics might be useful. Antiseptics, definitely. His med-kit—"That's all."

"Your blankets," said Hunt. "Y'want them, too."

Calhoun shrugged. He clothed himself for the cold outside. He had a blaster in his pocket, but he casually and openly took down a blast-rifle. His captors offered no objection. He shrugged again and replaced it. Starting to take it was only a test. He made a guess that this stocky leader, Hunt, might have kept his community just a little more nearly sane than the group that had set him to the cure of a sick cow. He hoped so.

"Murgatroyd," he said to the *tormal* still clinging up near the control-room's top, "we have a professional call to make. You'd better come along. In fact, you must."

Murgatroyd came suspiciously down, and then leaped to Calhoun's shoulder. He clung there, gazing distrustfully about. Calhoun realized that his captors—callers—whatever they were—stayed huddled away from every object in the cabin. They fingered nothing. But the scared eyes of most of them proved that it was not honesty which moved them to such meticulousness. It was fear. Of contagion.

"They're uncouth, eh?" said Calhoun sardonically. "But think, Murgatroyd, they may have hearts of gold! We physicians have to pretend to think so, in any case!"

"*Chee!*" said Murgatroyd resentfully as Calhoun moved toward the lock.

III

"Civilization is based upon rational thought applied to the purposes of men. Most mistakes occur in the process of thinking. But there can be a deep and fundamental error about purposes. It is simply a fact that the purposes of human beings are not merely those of rational animals. It is the profoundest of errors to believe otherwise—to consider, for example, that prosperity, or pleasure, or even survival, cannot be priced so high that their purchase is a mistake."

The Practice of Thinking
Fitzgerald

There was a sheet of paper fastened outside the combination lock of the Med Ship's exit port. It said that Calhoun had been taken away by men of Two City, to tend some sick person. It said that he would be returned. The latter part might not be believed, but the Med Ship might not be destroyed. The colony of the landing grid might try to break into it, but success was unlikely.

Meanwhile, it was an odd feeling to cross the grassy pastureland with hoarfrost crunching underfoot. The grid's steel girders made a harsh lace of blackness against the sky, with its shining ribbon slashing across it. But Calhoun found himself reflecting that the underground heat applied to the thousand-acre pasture had been regulated with discretion. There was surely power enough available from the grid to turn the area into a place of tropic warmth, in which only lush and thick-leaved vegetation could thrive. But a storm from the frigid mountains would destroy such plants. Hardy, low-growing, semi-arctic grass was the only suitable ground-cover. The iciest of winds could not freeze it so long as the ground was warmed.

Tonight's wind was biting. Calhoun had donned a parka of synthetic fur on which frost would not congeal at any temperature, but he was forced to draw fur before his face and adjust heated goggles before his eyes would stop

watering. Yet in the three-quarter-mile trudge to the edge of the snow, his feet became almost uncomfortably warm.

That, though, ended where a sledge waited at the edge of the snow. Five men had forced themselves inside the Med Ship. A sixth was on guard beside the sledge. There had been no alarm. Now the stocky man, Hunt, urged him to a seat upon the sledge.

"I'm reasonably able-bodied," said Calhoun mildly.

"You don't know where we're going—or how," growled Hunt.

Calhoun got on the sledge. The runners were extraordinarily long. He could not see small details, but it appeared that the sledge had been made of extreme length to bridge crevasses in a glacier. There were long thin metal tubes to help. At the same time, it looked as if it could be made flexible to twist and turn in a narrow or obstacle-strewn path.

The six clumsily-clad men pushed it a long way, while Calhoun frowned at riding. Then Murgatroyd shivered, and Calhoun thrust him inside the parka. There Murgatroyd wriggled until his nose went up past Calhoun's chin and he could sniff the outside air. From time to time he withdrew his nose—perhaps with frost-crystals on it. But always he poked his small black snout to sniff again. His whiskers tickled.

Two miles from the pastureland, the sledge stopped. One man fumbled somewhere behind Calhoun's seat and a roaring noise began. All six piled upon the long, slender snow-vehicle. It began to move. A man swore. Then, suddenly, the sledge darted forward and went gliding up a steep incline. It gathered speed. Twin arcs of disturbed snow rose up on either side, like bow-waves from a speeding water-skimmer. The sledge darted into a great ravine of purest white and the roaring sound was multiplied by echoes.

For better than half an hour, then, Calhoun experienced a ride which for thrills and beauty and hair-raising suspense made mere space-travel the stodgiest of transportation. Once the sledge shot out from beetling cliffs—all icy and glittering in the light from the sky—and hurtled down a slope of snow so swiftly that the wind literally whistled about the bodies of its occupants. Then the drive roared more loudly, and there was heavy deceleration, and abruptly the sledge barely crawled. The flexibility of the thing came into operation. Four of the crew, each controlling one segment of

the vehicle, caused it to twist and writhe over the surface of a glacier, where pressure-ridges abounded and pinnacles of shattered, squeezed-up ice were not uncommon.

Once they stopped short and slender rods reached out and touched, and the sledge slid delicately over them and was itself a bridge across a crevasse in the ice that went down unguessably. Then it went on and the rods were retrieved. Minutes later the sledge-motor was roaring loudly, but it barely crawled up to what appeared to be a mountain crest—there were ranges of mountains extending beyond seeing in the weird blue-and-golden skylight—and then there was a breathtaking dash and a plunge into what was incredibly a natural tunnel beside the course of an ice-river—and abruptly there was a vast valley below.

This was their destination. Some thousands of feet down in the very valley-bottom there was a strange, two-mile-long patch of darkness. The blue-gold light showed no color there, but it was actually an artificially warmed pastureland like that within and about the landing grid. But from this dark patch vapors ascended, and rolled, and gathered to form a misty roof—which was swept away and torn to tatters by an unseen wind.

The sledge slowed and stopped beside a precipitous upcrop of stone while still high above the valley bottom. A voice called sharply:

"It's us," growled Hunt's deep voice. "We got him. Everything all right?"

... "No!" rasped the invisible voice. "They broke out—he broke out and got her loose, and they run off again. We shoulda killed 'em and had done with it!"

Everything stopped. The men on the sledge seemed to become still in the shock of pure disaster, pure frustration. Calhoun waited. Hunt was motionless. Then one of the men on the sledge spat elaborately. Then another stirred.

"Had your work for nothing," rasped the voice from the shadow. "The trouble that's started goes for nothing, too!"

Calhoun asked crisply:

"What's this? My special patients ran away?"

"That the Med man we heard about?" The invisible speaker was almost derisive with anger. "Sure! They've run off, all right! Man and girl together. After we made trouble with Three City by not killin' 'em and One City by sneakin' over to get you! Three City men'll come boiling over—" The voice raised in pitch, expressing scorn and fury. "Because they fell in love! We shoulda killed 'em right off or let 'em die in the snow like they wanted in the first place!"

Calhoun nodded almost imperceptibly to himself. When there is a syndrome forbidding association between societies, it is a part of the society's interior struggle against morbidity that there shall be forbidden romances. The practice of exogamy is necessary for racial health, hence there is an instinct for it. The more sternly a small population restricts its human contacts to its own members, the more repressed the exogamic impulse becomes. It is never consciously recognized for what it is. But especially when repressed, other-than-customary contacts trigger it explosively. The romantic appeal of a stranger is at once a wise provision of nature and a cause of incredible furies and disasters. It is notorious that spaceship crews are inordinately popular where colonies are small and strangers infrequent. It is no less notorious that a girl may be destitute of suitors on her own world, but has nearly her choice of husbands if she merely saves the ship fare to another.

Calhoun could have predicted defiances of tradition and law and quarantine alike, as soon as he began to learn the state of things here. The frenzied rage produced by this specific case was normal. Some young girl must have loved terribly, and some young man been no less impassioned, to accept expulsion from society on a world where there was no food except in hydroponic gardens and artificially warmed pastures. It was no less than suicide for those who loved. It was no less than a cause for battle among those who did not.

The deep-voiced Hunt said now, in leaden, heavy tones:

"Cap it. This is my doing. It was my daughter I did it for. I wanted to keep her from dying. I'll pay for trying. They'll be satisfied in Three City and in One alike if you tell 'em it's my fault and I've been drove out for trouble-making."

Calhoun said sharply:

"What's that? What's going on now?"

The man in the shadows answered—by his tone as much to express disgust as to give information.

"His daughter Nym was on sentry-duty against Three City sneaks. They had a sentry against us. The two of 'em talked across the valley between 'em. They had walkies to report with. They used 'em to talk. Presently she sneaked a vision screen out of store. He prob'ly did, too. So presently they figured it was worth dyin' to die together. They run off for the hotlands. No chance to make it, o'course!"

The hotlands could hardly be anything but the warm equatorial belt of the planet.

"We should've let them go on and die," said the stocky Hunt, drearily, "but I persuaded men to help me bring 'em back. We were careful against sickness! And we ... I ... locked them separate and I ... I hoped my daughter mightn't die of the Three City sickness. I even hoped that young man wouldn't die of the sickness they say we have that we don't notice and they die of. Then we heard your call to One City. We couldn't answer it, but we heard all you said, even to the bargain about the cow. And ... we'd heard of Med men who cured sickness. I ... hoped you could save Nym from dying of the Three City sickness or passing it in our city. My friends risked much to bring you here. But ... my daughter and the man have fled again."

"And nobody's goin' to risk any more!" rasped the voice from the shadow of the cliff. "We held a council! It's decided! They're gone and we got to burn out the places they was in! No more! You don't head the Council any more, either! We decided that, too. And no Med man! The Council ruled it!"

Calhoun nodded yet again. It is a part of fear, elaborately to ignore everything that can be denied about the thing feared. Which includes rational measures against it. This was a symptom of the state of things which constituted a Med Service emergency, because it caused needless deaths.

Hunt made a gesture which was at once commanding and filled with despair.

"I'll take the Med man back so One City can use him if they dare and not blame you for me taking him. I'll have to take the sledge—but he's used it so it'd have to be burned anyhow. You men be sure to burn your clothes. Three

City'll be satisfied because I'm lost to balance for their man lost. The Med man will tell One City I'm drove out. You've lost me and my daughter too, and Three City's lost a man. One City'll growl and threaten, but they win by this. They won't risk a showdown."

Silence again. As if reluctantly, one man of the party that had abducted Calhoun moved away from the sledge and toward the abysmally deep shadow of the cliff. Hunt said harshly:

"Don't forget to burn your clothes! You others, get off the sledge. I'm taking the Med man back and there's no need for a war because I made the mistake and I'm paying for it."

The remaining men of the kidnaping-party stepped off the sledge into the trampled snow, just here. One said clumsily:

"Sorry, Hunt. Luck!"

"What luck could I have?" asked the stocky man, wearily.

The roaring of the sledge's drive, which had been a mere muffled throbbing, rose to a booming bellow. The snow-vehicle surged forward, heading downward into the valley with the dark area below. Half a mile down, it began to sweep in a great circle to return upon its former track. Calhoun twisted in his seat and shouted above the roar. He made violent gestures. The deep-voiced Hunt, driving from a standing position behind the seat, slowed the sledge. It came nearly to a stop and hissing noises from snow passing beneath it could be heard.

"What's the matter?" His tone was lifeless. "What d'you want?"

"Two people have run away," said Calhoun vexedly. "Your daughter Nym and a man from Three City—whatever that is. You're driven out to prevent fighting between the cities."

"Yes," said Hunt, without expression.

"Then let's go get the runaways," said Calhoun irritably, "before they die in the snow! After all, you got me to have me save them! And there's no need for anybody to die unless they have to!"

Hunt said without any expression at all:

"They're heading for the hotlands—where they'd never get. It's my meaning to take you back to your ship, and then find them and give them the sledge so's they'll ... so Nym will keep on living a while longer."

He moved to shift the controls and set the sledge again in motion. His state of mind was familiar enough to Calhoun—shock or despair so great that he could feel no other emotion. He would not react to argument. He could not weigh it. He'd made a despairing conclusion and he was lost to all thought beyond carrying it out. His intention was not simply a violent reaction to a single event, such as an elopement. He intended desperate means by which a complex situation could be kept from becoming a catastrophe to others. Three City had to be dealt with in this fashion, and One City in that, and it was requisite that he die, himself. Not only for his daughter but for his community. He had resolved to go to his death for good and sufficient reasons. To get his attention to anything else, he would have to be shocked into something other than despair.

Calhoun brought his hand out of its pocket. He held a blaster. He'd pocketed the weapon before he went to examine the cow. He'd had the power to stop his own abduction at any instant. But a medical man does not refuse a call for professional service.

Now he pointed the blaster to one side and pressed the stud. A half-acre of snow burst into steam. It bellowed upward and went writhing away in the peculiar blue-gold glow of this world at night.

"I don't want to be taken back to my ship," said Calhoun firmly. "I want to catch those runaways and do whatever's necessary so they won't die at all. The situation here has been thrown into my lap. It's a Med Service obligation to intervene in problems of public health, and there's surely a public-health problem here!"

Murgatroyd wriggled vigorously under Calhoun's parka. He'd heard the spitting of the blaster and the roaring of exploded steam. He was disturbed. The stocky man stared.

"What's that?" he demanded blankly. "You pick up—"

"We're going to pick up your daughter and the man she's with," Calhoun told him crossly. "There's an isolation syndrome and what looks like a Crusoe

problem here! It's got to be dealt with! As a matter of public health!"

The stocky Hunt started at him. Calhoun's intentions were unimaginable to him. He floundered among incredible ideas.

"We medics," said Calhoun, "made it necessary for men to invent interplanetary travel because we kept people from dying and the population on old Earth got too large. Then we made interstellar travel necessary because we continued to keep people from dying and one solar system wasn't big enough. We're responsible for nine-tenths of civilization as it exists today, because we produced the conditions that make civilization necessary! And since on this planet civilization is going downhill and people are dying without necessity, I have the plain obligation to stop it! So let's go pick up your daughter Nym and this sweetheart of hers, and keep them from dying and get civilization on the upgrade again!"

The former leader of the kidnapers said hoarsely:

"You mean—" Then he stammered. "Th-th-they're heading for the hotlands. No other way to go. Watch for their tracks!"

The drive-engine bellowed. The sledge raced ahead. And now it did not complete the circle that had been begun, to head back to the landing grid. Now it straightened and rushed in a splendid roaring fierceness down between the sides of the valley. It left behind the dark patch with its whirling mists. It flung aside bow-waves of fine snow, which made rainbows in the half-light which was darkness here. It rushed and rushed and rushed, leaving behind a depression which was a singular permanent proof of its passage.

Calhoun cringed a little against the wind. He could see little or nothing of what was ahead. The sprayed wings of upflung snow prevented it. Hunt, standing erect, could do better. Murgatroyd, inside the parka, again wriggled his nose out into the stinging wind and withdrew it precipitately.

Hunt drove as if confident of where to go. Calhoun dourly began to fit things into the standard pattern of how such things went. There were self-evidently three cities or colonies on this planet. They'd been named and he'd seen three patches of pasture from the stratosphere. One was plainly warmed by power

applied underground—electric power from the landing grid's output. The one now falling behind was less likely to be electrically heated. Steam seemed more probable because of the vapor-veil above it. This sledge was surely fuel-powered. At a guess, a ram-jet drove it. Such motors were simple enough to make, once the principle of air inflow at low speeds was known. Two City—somewhere to the rear—might operate on a fuel technology which could be based on fossil oil or gas. The power-source for Three City could not now be guessed.

Calhoun scowled as he tried to fill in the picture. His factual data was still limited. There was the misty golden ribbon in space. It was assuredly beyond the technical capacity of cities suffering from an isolation syndrome. He'd guessed at hydroponic gardens underground. There was surely no surface city near the landing grid, and the city entrance they'd just left was in the face of a cliff. Such items pointed to a limited technical capacity. Both, also, suggested mining as the original purpose of the human colony or colonies here.

Only mining would make a colony self-supporting in an arctic climate. This world could have been colonized to secure rare metals from it. There could be a pipeline from an oil field or from a gas well field near a landing grid. Local technological use of gas or oil to process ores might produce ingots of rare metal worth interstellar freight charges. One could even guess that metal reduced by heat-chemistry could be transported in oil suspension over terrain and under conditions when other forms of surface transportation were impractical.

If the colony began as a unit of that sort, it would require only very occasional visits of spacecraft to carry away its products. It could be a company-planet, colonized and maintained by a single interstellar corporation. It could have been established a hundred and fifty or two hundred years before, when the interstellar service organizations were in their infancy and only operated where they were asked to serve. Such a colony might not even be on record in the Medical Service files.

And that would account for everything. When for some reason the mines became unprofitable, this colony would not be maintained. The people who wished to leave would be taken off—of course. But some would elect to stay behind in the warmed, familiar cities they and their fathers had been born in. They couldn't imagine moving to a strange and unfamiliar world.

So much was normal reasoning. Now the strictly technical logic of the Med Service took over to explain the current state of things. In one century or less an isolated community could lose, absolutely, its defenses against diseases to which it was never exposed. Amerinds were without defense against smallpox, back on Earth. A brown race scattered among thousands of tiny islands was nearly wiped out by measles when it was introduced. Any contact between a long-isolated community and another—perhaps itself long-isolated—would bring out violently any kind of contagion that might exist in either.

There was the mechanism of carriers. The real frequency of disease-carriers in the human race had been established less than two generations ago. A very small, isolated population could easily contain a carrier or carriers of some infection. They could spread it so freely that every member of their group acquired immunity during infancy. But a different isolated group might contain a carrier of a different infection and be immune but distributive of it.

It was literally true that each of the three cities might have developed in their first century of isolation a separate immunity to one disease and a separate defenselessness against all others. A member of one community might be actually deadly to a member of either of the others whom he met face to face.

With icy wind blowing upon him as the sledge rushed on, Calhoun wryly realized that all this was wholly familiar. It was taught, nowadays, that something of the sort had caused the ancient, primitive human belief that women were perilous to men, and a man must exercise great precaution to avoid evil *mana* emanating from his prospective bride. When wives were acquired by capture and all human communities were small and fiercely self-isolated—why each unsanitary tribal group might easily acquire a condition like that assumed in cities One, Two, and Three. The primitive suspicion of woman would have its basis in reality if the women of one tribe possessed immunity to some deadly microbe their skin or garments harbored—and if their successful abductors had no defense against it.

The speeding sledge swerved. It leaned inward against the turn. It swerved again, throwing monstrous sheets of snow aloft. Then the drive-jet lessened its roar. The shimmering bow-waves ceased. The sledge slowed to a mere headlong glide.

"Their trail!" Hunt cried in Calhoun's ear.

Calhoun saw depressions in the snow. There were two sets of pear-shaped dents in the otherwise virgin surface. Two man beings, wearing oblong frames on their feet, criss-crossed with cordage to support them atop the snow, had trudged ahead, here, through the gold-blue night.

Calhoun knew exactly what had happened. He could make the modifications the local situation imposed upon a standard pattern, and reconstitute a complete experience leading up to now.

A girl in heavy, clumsy garments had mounted guard in a Two City sentry-post above a snow-filled mountain valley. There were long and bitter-cold hours of watching, in which nothing whatever happened. Eternal snows seemed eternally the same, and there was little in life but monotony. But she'd known that across the valley there was another lonely watcher from an alien city, the touch of whose hand or even whose breath would mean sickness and death. She'd have mused upon the strangeness that protected her in this loneliness—because her touch or her breath would be contagion upon him, too. She'd have begun by feeling a vague dread of the other sentry. But presently, perhaps, there came a furtive call on the walkie-frequency used by sentries for communication with their own cities.

Very probably she did not answer at first. But she might listen. And she would hear a young man's voice, filled with curiosity about the sentry who watched as he did.

There'd come a day when she'd answer shyly. And there would be relief and a certain fascination in talking to someone so much like herself—but so alien and so deadly! Of course there could be no harm in talking to someone who would flee from actual face-to-face contact as desperately as herself. They might come to joke about their mutual dangerousness. They might find it amusing that cities which dared not meet should hate. Then there'd come a vast curiosity to see each other. They'd discuss that frankly—because what possible evil could come, if two persons were deadly to each other should they actually approach?

Then there'd come a time when they looked at each other breathlessly in vision-screens they'd secretly stolen from their separate cities' stores. There could be no harm. They were only curious! But she would see someone at once infinitely strange but utterly dear, and he would see someone lovely beyond the girls of his own city. Then they would regret the alienness which made them perilous to each other. Then they would resent it fiercely. They'd end by denying it.

So across the wide valley of eternal snow there would travel whispers of desperate rebellion, and then firmly resolute murmurings, and then what seemed the most obvious of truths—that it would be much more satisfactory to die together than to live apart. And insane plannings would follow—arrangements by which two trembling young folk would meet secretly and flee. Toward the hotlands, to be sure, but without any belief than that the days before death, while they were together, were more precious than the lifetimes they would give up to secure them.

Calhoun could see all this very clearly, and he assured himself that he regarded it with ironic detachment. He asserted in his own mind that it was merely the manifestation of that blind impulse to exogamy which makes spacemen romantic in far spaceports and invests an outer-planet girl with glamour. But it was something more. It was also that strange and unreasonable and solely human trait which causes one to rejoice selflessly that someone else exists, so that his or her own life and happiness is put into its place of proper insignificance in the cosmos. It may begin in instinct, but it becomes an achievement only humans can encompass.

Hunt knew it—the stocky, deep-voiced despairing figure who stared hungrily for the daughter who had defied him and for whom he was an exile from all food and warmth.

He flung out a mittened hand.

"There!" he cried joyously. "It's them!"

There was a dark speck in the blue-gold night-glow. As the sledge swept close, there were two small figures who stood close together. They defiantly faced the approaching sledge. As its drive-motor stopped and it merely glided on, its runners whispering on the snow, the girl snatched away the cold-mask which all the inhabitants of this planet wore out-of-doors. She raised her face to the man. They kissed.

And then the young man desperately raised a knife. It glittered in the light of the ribbon in the sky. And—

Calhoun's blaster made its inadequate rasping noise. The knife-blade turned incandescent for two-thirds of its length. The young man dropped the suddenly searing handle. The knife sank hissing into the snow.



"It's always thrilling to be dramatic," said Calhoun severely, "but I assure you it's much more satisfying to be sane. The young lady's name is Nym, I believe. I do not know the gentleman. But Nym's father and myself have come to put the technical resources of two civilizations at your disposal as a first step toward treatment of the pandemic isolation syndrome on this planet, which with the complications that have developed amounts to a Crusoe health problem."

Murgatroyd tried feverishly to get his head out of Calhoun's parka past his chin. He'd heard a blaster. He sensed excitement. His nose emerged, whiffing

frantically. Calhoun pushed it back.

"Tell them, Hunt," he said irritably. "Tell them what we're here for and what you've done already!"

The girl's father told her unsteadily—almost humbly, for some reason—that the jet-sledge had come to take her and her sweetheart—to be her husband—to the hotlands where at least they would not die of cold. Calhoun added crossly that he believed there would even be food there—because of the ribbon in the sky.

Trembling and abashed, the fugitives got on the sledge. Its motor roared. It surged toward the hotlands under the golden glow of that ribbon—which obviously had no rational explanation unless somebody had made a grave mistake. But Calhoun had not.

IV

"An action is normally the result of a thought. Since we cannot retract an action, we tend to feel that we cannot retract the thought which produced it. In effect, we cling desperately to our mistakes. In order to change our views we have commonly to be forced to act upon new thoughts, so urgent and so necessary that without disowning our former, mistaken ideas, we can abandon them tactfully without saying anything to anybody—even ourselves."

The Practice of Thinking
Fitzgerald

Murgatroyd came down a tree with his cheek-pouches bulged with nuts. Calhoun inserted a finger, and Murgatroyd readily permitted him to remove and examine the results of his scramble aloft. Calhoun grunted. Murgatroyd did have other and more useful abilities in the service of public health, but right here and now his delicate digestion was extremely convenient. His stomach worked so much like a human's, that anything Murgatroyd ate was safe for Calhoun to an incredible degree of probability. And Murgatroyd ate nothing that disagreed with him.



"Instead of 'physician, heal thyself,'" Calhoun observed, "it's amounted to 'physician, feed thyself,' since we got past the frost-line, Murgatroyd. I am gratified."

"*Chee!*" said Murgatroyd complacently.

"I expected," said Calhoun, "only to benefit by the charm of your society in what I thought would be a routine check-trip to Merida Two. Instead, some unknown fumble-finger punched a wrong button and we wound up here. Not exactly here, but near enough. I brought you from the Med Ship because there was nobody to stay around and feed you, and now you feed us—at least by pointing out edible things we might otherwise miss."

"*Chee!*" said Murgatroyd. He strutted.

"I wish," protested Calhoun annoyedly, "that you wouldn't imitate that Pat character from Three City! As a brand-new husband he's entitled to strut a

little, but I object to your imitating him! You haven't anybody acting like Nym!—gazing at you raptly as if you'd invented not only marriage but romance itself, impassioned falsehoods, and all other desirable things back to night and morning!"

Murgatroyd said, "*Chee?*" and turned to face away from Calhoun.

The two of them, just then, stood on a leaf-covered patch of ground which slanted down to the singularly smooth and reflective water of a tiny bay. Behind and above them reared gigantic mountains. There was snow in blinding-white sheets overhead, but the snowline itself was safely three thousand feet above them. Beyond the bay was a wide estuary, with more mountains behind it, with more snowfields on their flanks. A series of leaping cascades jumped downward from somewhere aloft where a glacier-foot melted in the sun's heat. And everywhere that snow was not, green stuff shone in the sunlight.

Nym's father, Hunt, came hurriedly toward the pair. He'd abandoned the thick felt cloak and heavy boots of Two City. Now he was dressed nearly like a civilized man, but he carried a sharpened stick in one hand and in the other a string of authentic fish. He wore an expression of astonishment. It was becoming habitual.

"Murgatroyd," said Calhoun casually, "has found another kind of edible nut. Terrestrial, too, like half the living things we've seen. Only the stuff crowding the glaciers seems to be native. The rest originated on Earth and was brought here, some time or another."

Hunt nodded. He seemed to find some difficulty in speaking.

"I've been talking to Pat," he said at last.

"The son-in-law," observed Calhoun, "who has to thank you not only for your daughter and his life, but for your public career in Two City which qualified you to perform a marriage ceremony. I hope he was respectful."

Hunt made an impatient gesture.

"He says," he protested, "that you haven't done anything either to Nym or to him to keep them from dying!"

Calhoun nodded.

"That's true."

"But ... they should die! Nym should die of the Three City sickness! And Three City people have always said that we had a sickness too ... that did not harm us but they died of!"

"Which," agreed Calhoun, "is undoubtedly historical fact. Its current value is that of one factor in an isolation syndrome and consequently a complicating factor in the Crusoe health problem here. I've let Nym and Pat go untreated to prove it. I think there's only a sort of mass hypochondria based on strictly accurate tradition. Which would be normal."

Hunt shook his head.

"I don't understand!" he protested helplessly.

"Someday I'll draw a diagram," Calhoun told him. "It is complicated. Did you check with Pat on what Three City knows about the ribbon in the sky? I suspect it accounts for the terrestrial plants and animals here, indirectly. There wouldn't be an accidental planting of edible nuts and fish and squirrels and pigeons and rabbits and bumblebees! I suspect there was a mistake somewhere. What does Pat say?"

Hunt shrugged his shoulders.

"When I talk to him," added Calhoun, "he doesn't pay attention. He simply gazes at Nym and beams. The man's mad! But you're his father-in-law. He has to be polite to you!"

Hunt sat down abruptly. He rested his spear against a tree and looked over his string of fish. He wasn't used to the abundance of foodstuffs here, and the temperature—Calhoun estimated it at fifty degrees—seemed to him incredibly balmy. Now he thoughtfully separated one fish from the rest and with a certain new skill began to slice away two neatly boneless fillets. Calhoun had showed him the trick the day after a lesson in fish-spearing, which was two days after their arrival.

"Children in Three City," growled Hunt, "are taught the same as in Two City. Men came to this planet to work the mines. There was a company which sent them, and every so often it sent ships to take what the mines yielded, and to bring things the people wanted. Men lived well and happily. The company

hung the ribbon in the sky so the hotlands could grow food for the men. But presently the mines could not deliver what they made to the ships when they came. The hotlands grew bigger, the glaciers flowed faster, and the pipes between the cities were broken and could not be kept repaired. So the company said that since the mine-products could no longer be had, it could not send the ships. Those who wanted to move to other worlds would be carried there. Some men went, with their wives and children. But the grandfathers of our fathers' grandfathers were contented here. They had homes and heat and food. They would not go."

Hunt regarded the pinkish brook trout fillet he'd just separated. He bit off a mouthful and chewed, thoughtfully.

"That really tastes better cooked," said Calhoun mildly.

"But it is good this way also," said Hunt. He was grizzled and stocky and somehow possessed of dignity which was not to be lost merely by eating raw fish. He waved the remainder of the fillet. "Then the ships ceased to come. Then sickness came. One City had a sickness it gave to people of Two and Three when they visited it. Two City had a sickness it gave to One and Three. Three City—" He grunted. "Our children in Two say only Two City people have no sickness. Three City children are taught that only Three City is clean of sickness."

Calhoun said nothing. Murgatroyd tried to gnaw open one of the nuts he'd brought down from the tree. Calhoun took it and another and struck them together. Both cracked. He gave them to Murgatroyd, who ate them with great satisfaction.

Hunt looked up suddenly.

"Pat did not give a Three City sickness to Nym," he observed, "so our thinking was wrong. And Nym has not given a Two City sickness to him. His thinking was wrong."

Calhoun said meditatively:

"It's tricky. But sickness can be kept by a carrier, just as you people have believed of other cities. A carrier has a sickness but does not know it. People around the carrier have the sickness on their bodies or their clothing from the carrier. They distribute it. Soon everybody in the city where there is a carrier—" Calhoun had a moment's qualm because he used the word "city." But to

Hunt the idea conveyed was a bare few hundred people. "Soon everybody is used to the sickness. They are immune. They cannot know it. But somebody from another city can come, and they are not used to the sickness, and they become ill and die."

Hunt considered shrewdly.

"Because the sickness is on clothing? From the carrier?"

Calhoun nodded.

"Different carriers have different sicknesses. So one carrier in One City might have one disease, and all the people in One City became used to it while they were babies—became immune. There could be another carrier with another sickness in Two City. A third in Three City. In each city they were used to their own sickness—"

"That is it," said Hunt, nodding. "But why is Pat not dying? Or Nym? Why do you do nothing to keep them alive?"

"Suppose," said Calhoun, "the carrier of a sickness dies. What happens?"

Hunt bit again, and chewed. Suddenly he choked. He sputtered:

"There is no sickness to spread on the clothing! The people no longer have it to give to strangers who are not used to it! The babies do not get used to it while they are little! There is no longer a One City sickness or a Two City sickness or a Three!"

"There is," said Calhoun, "only a profound belief in them. You had it. Everybody else still has it. And the cities are isolated and put out sentries because they believe in what used to be true. And people like Nym and Pat run away in the snow and die of it. There is much death because of it. You would have died of it."

Hunt chewed and swallowed. Then he grinned.

"Now what?" His deep voice was quaintly respectful to Calhoun, so much younger than himself. "I like this! We were not fools to believe, because it was true. But we are fools if we still believe, because it is not true any more. How do we make people understand, Calhoun? You tell me. I can handle people when they are not afraid. I can make them do what I think wise—when they are not afraid. But when they fear—"

"When they fear," said Calhoun dryly, "they want a stranger to tell them what to do. You came for me, remember? You are a stranger to One City and Three City. Pat is a Stranger to Two City. If the cities become really afraid —"

Hunt grunted. He watched Calhoun intently. And Calhoun was peculiarly reminded of the elected president of a highly cultured planet, who had exactly that completely intent way of looking at one.

"Go on!" said Hunt. "How frighten them into—this?"

He waved his hand about. Calhoun, his tone very dry indeed, told him. Words would not be enough. Threats would not be enough. Promises would not be enough. But rabbits and pigeons and squirrels and fish—fish that were frozen like other human food—and piles of edible nuts.... They would not be enough either, by themselves. But—

"An isolation syndrome is a neurotic condition, and a Crusoe problem amounts to neurotic hypochondria. You can do it—you and Pat."

Hunt grimaced.

"I hate the cold, now. But I will do it. After all, if I am to have grandchildren there should be other children for them to play with! And we will take you back to your ship?"

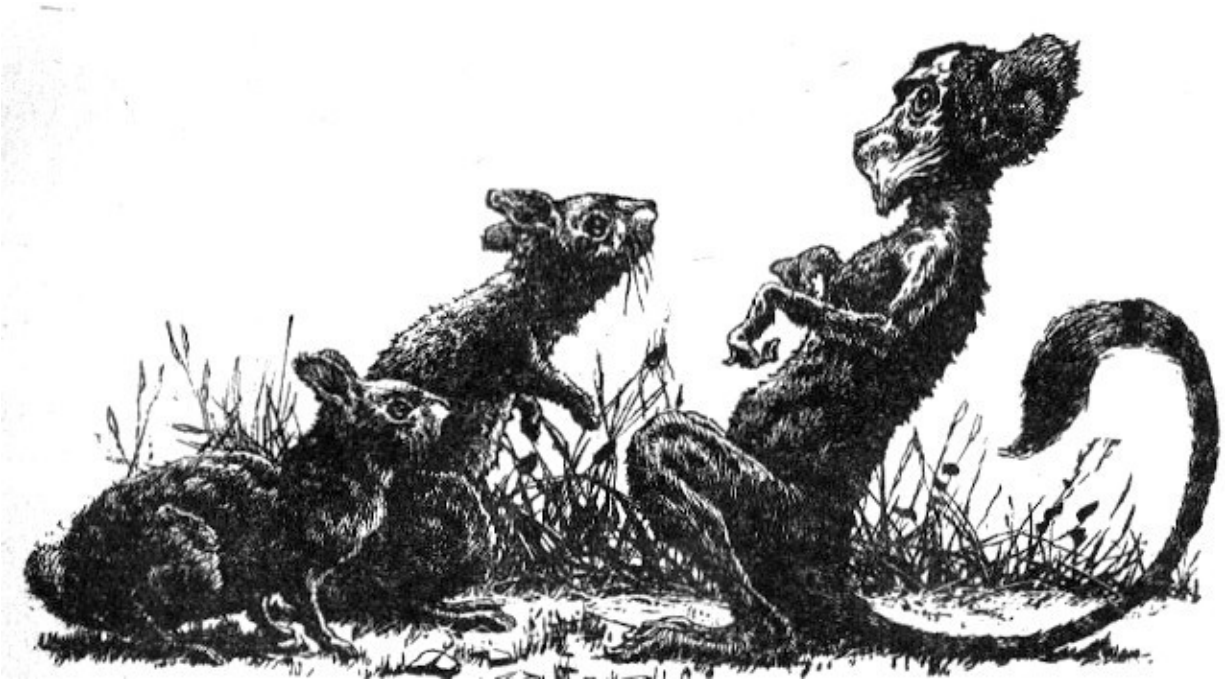
"You will," said Calhoun. "By the way, what is the name of this planet, anyhow?"

Hunt told him.

Calhoun slipped across the pasture inside the landing grid and examined the ship from the outside. There had been batterings, but the door had not been opened. In the light of the ribbon in the sky he could see, too, that the ground was trampled down but only at a respectful distance. One City was disturbed

about the Med Ship. But it did not know what to do. So long as nothing happened from it....

He was working the combination lock-door when something hopped, low-down and near him. He jumped, and Murgatroyd said, "*Chee?*" Then Calhoun realized what had startled him. He finished the unlocking of the port. He went in and closed the port behind him. The air inside seemed curiously dead, after so long a time outside. He flipped on the outside microphones and heard tiny patterings. He heard mildly resentful cooings. He grinned.



When morning came, the people of One City would find their pastureland inhabited by small snowshoe rabbits and small and bush-tailed squirrels and fluttering pigeons. They would react as Two City and Three City had already done—with panic. And panic would inevitably call up the notion of the most-feared thing in their lives. Sickness. The most-feared thing is always a rare thing, of course. One cannot fear a frequent thing, because one either dies of it or comes to take it for granted. Fear is always of the rare or nonexistent. One City would be filled with fear of sickness.

And sickness would come. Hunt would call them, presently, on a walkie-talkie communicator. He would express deep concern because—so he'd say—new domestic animals intended for Two City had been dumped on One City pastureland. He'd add that they were highly infective, and Two City was

already inescapably doomed to an epidemic which would begin with severe headaches, and would continue with cramps and extreme nervous agitation. And he would say that Calhoun had left medicines at Two City with which that sickness and all others could be cured, and if the sickness described should appear in One City—why—its victims could be cured if they traveled to Two City.

The sickness would appear. Inevitably. There was no longer sickness in the three communities. Arctic colonies, never visited by people from reservoirs of infection, become magnificently healthy by the operation of purely natural causes. But an isolation syndrome....

The people of One City would presently travel, groaning, to Two City. Their suffering would be real. They would dread the breaking of their isolation. But they'd dread sickness—even sickness they only imagined—still more. And when they reached Two City they would find themselves tended by Three City members, and they would be appalled and terrified. But mumbo-jumbo medication by Hunt and Pat—and Nym for the women—would reassure them. A Crusoe condition requires heroic treatment. This was it.

Calhoun cheerfully checked over the equipment of the Med Ship. He'd have to take off on emergency rockets. He'd have to be very, very careful in setting a course back to Headquarters to report before starting out again for Merida II. He didn't want to make any mistakes. Suddenly he began to chuckle.

"Murgatroyd," he said amiably, "it's just occurred to me that the mistakes we make—that we struggle so hard to avoid—are part of the scheme of things."

"*Chee?*" said Murgatroyd inquiringly.

"The company that settled this planet," said Calhoun, grinning, "set up that ribbon out in space as a splendidly conservative investment to save money in freight charges. It was a mistake, because it ruined their mining business and they had to write the whole colony off. They made another mistake by not reporting to Med Service, because now they've abandoned the colony and would have to get a license to re-occupy—which they'd never be granted against the population already there. Somebody made a mistake that brought

us here, and One City made a mistake by not accepting us as guests, and Two City made a mistake by sending Nym on sentry-duty, and Three City made a mistake...."

Murgatroyd yawned.

"You," said Calhoun severely, "make a mistake in not paying attention." He strapped himself in. He stabbed an emergency-rocket control-button. The little ship shot heavenward on a pencil-thin stream of fire. Below him, people of One City would come pouring out of underground to learn what had happened, and they'd find the pasture swarming with friendly squirrels and inquisitive rabbits and cooing pigeons. They'd be scared to death. Calhoun laughed. "I'll spend part of the time in overdrive making a report on it. Since an isolation syndrome is mostly psychological, and a Crusoe condition is wholly so—I managed sound medical treatment by purely psychological means! I'll have fun with that!"

It was a mistake. He got back to Headquarters all right, but when his report was read they made him expand it into a book, with footnotes, an index, and a bibliography.

It was very much of a mistake!

THE END

*** END OF THE PROJECT GUTENBERG EBOOK RIBBON IN THE
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